Option A shows an **exponential decay of power with a constant half-life**.

25 C The magnetic force acting on the proton is upwards. To balance this force, so that the proton would pass undeflected, the electric force on the proton acts downwards, so the electric field is directed downwards.

$$Bqv = qE$$

$$v = \frac{E}{B} \rightarrow E = Bv = (0.015)(2.0 \times 10^7) = 3.0 \times 10^5 \text{ N C}^{-1}$$

Note that the sign and magnitude of the charge do not matter; if the speed is right, the particle is undeflected. If it were a negative charge the electric force and magnetic force directions just swops.